

Securing Big Data: A Multi-Layered Approach

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2. ABSTRACT:

- Big data systems are used in finance, healthcare and cloud platforms, where systems face major challenges in data security, privacy and compliance.
- A multi-layer security framework that integrates:
 - AES-256 encryption
 - SHA-256 hashing
 - Access control mechanisms
 - Compliance models
- This framework ensure confidentiality, integrity, and availability (CIA) with minimal performance overhead.



3. INTRODUCTION:

Big data in finance is highly sensitive, and any security breach can lead to financial loss, trust deficit, and system instability.

However, due to High data volume, Real-time processing, Sensitive nature of financial data

These systems become highly vulnerable to Data breaches, Unauthorized access, Data tampering

This creates a strong need for a **robust and integrated security solution**, which motivated our research.



4. LITERATURE REVIEW:

Existing research focuses on individual security techniques:

- **Encryption:** RSA, AES, Hybrid methods
- **Access Control:** RBAC and ABAC
- **Compliance Models:** GDPR, HIPAA

However, most of these approaches are **standalone** and lack integration, making them insufficient for real-time financial systems.



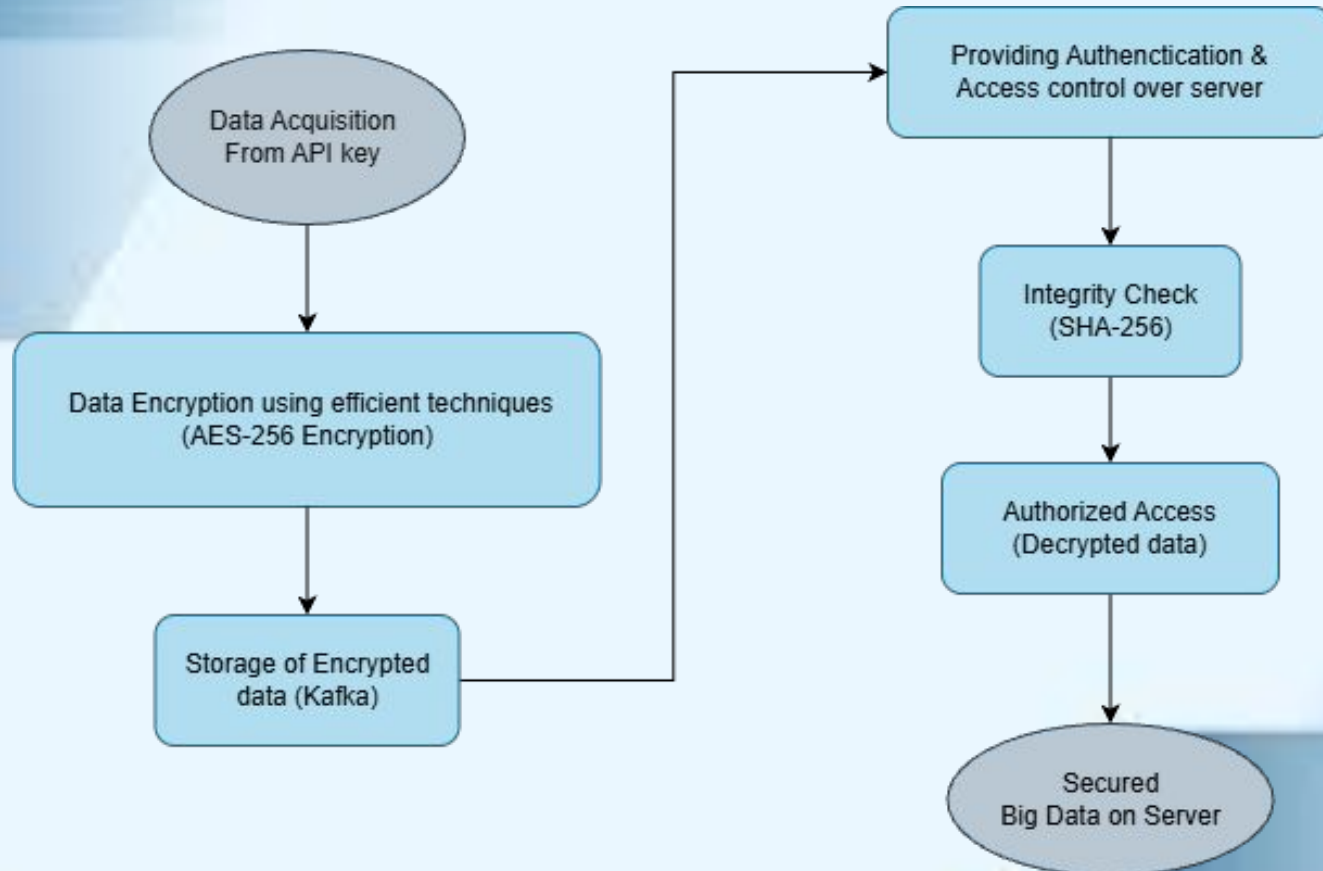
6. RESEARCH GAPS:

From our study, we identified key gaps:

- Lack of **integrated security frameworks**
- Weak **real-time protection mechanisms**
- Limited **practical compliance enforcement**
- Poor scalability under high data loads



6. PROPOSED METHODOLOGY :




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Pretty-print ☐

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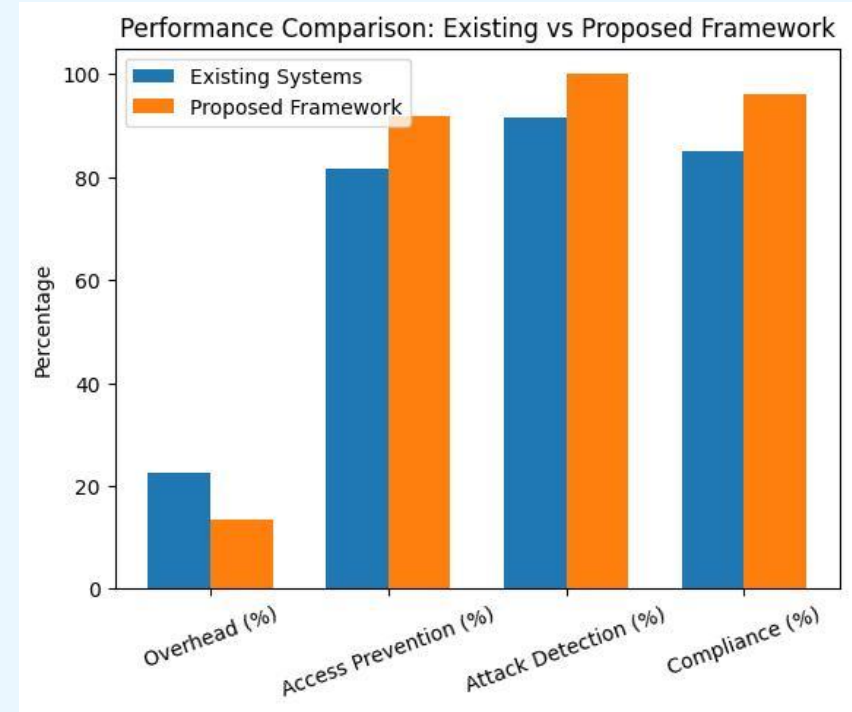
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Link to API:

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7. RESULTS & DISCUSSION:

- AES-256 encryption introduced only **12–15% computational overhead**, which remained stable even under high data loads
- **SHA-256 hashing successfully detected all data tampering attempts** with zero false negatives
- **RBAC + ABAC access control prevented ~92% of unauthorized access attempts**
- **Compliance module identified ~96% of GDPR and HIPAA violations**
- System maintained **high throughput and stability** using Kafka and Hadoop under real-time conditions



8. COMPARATIVE ANALYSIS:

Aspect	Prior Works	Proposed Framework
Encryption	RSA / Hybrid (slow, complex)	AES-256 (secure, efficient, 12–15% overhead)
Integrity	Weak / missing	SHA-256 detects all tampering
Access Control	Mostly RBAC	RBAC + ABAC (>90% attacks blocked)
Compliance	Theoretical (GDPR/HIPAA only)	Practical enforcement with logging & audit
Performance	Limited real-time validation	Stable under Kafka & Hadoop workloads
Scalability	Struggles with large data loads	Smooth scaling in big data environments
Coverage	Partial solutions	End-to-end security framework

9. CONCLUSION & FUTURE WORK :

- Big data improves financial decision-making but introduces serious security risks.
- Proposed **multi-layered security framework** ensures:
 - Confidentiality (AES-256)
 - Integrity (SHA-256)
 - Secure transfer (SSH)
 - Controlled access
- Achieves **high security with minimal performance overhead**
- Suitable for **real-time financial systems**
- Future scope:
 - AI-based threat detection
 - Quantum-resistant cryptography
 - Enhanced scalability

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Thank you